

PROJECT PRESENTATION & CREDENTIAL OVERVIEW

Machine Learning Internship

Arsalan Adil

Roll No & Intern ID

**Roll No: 25SCS1003003061 • ID:
CTI/C1/B903531**

Academic & Organization Track

**B.Tech CSE • IILM University | Cognifyz IT Solutions (06/07/2026 -
06/08/2026)**

Internship Overview

A high-level summary of the machine learning track completed at Cognifyz IT Solutions Pvt. Ltd.



Applied ML Practice

Hands-on end-to-end model development pipelines using standard data science ecosystems.



Real-World Datasets

Predictive modeling across venue metadata, rating forecasts, and customer preference analytics.



Verified Credentials

Successfully completed all four specialized project modules with verified organization endorsement.

Candidate & Program Details

Academic affiliation and industry internship verification details.

Intern Credentials

Name: Arsalan Adil

Roll Number: 25SCS1003003061

Intern ID: CTI/C1/B903531

Academic Track: B.Tech CSE (Batch 2025-2029)

University: IILM University, Greater Noida

Organization & Tenure

Company: Cognifyz IT Solutions Pvt. Ltd.

Domain Track: Machine Learning

Tenure Duration: 06/07/2026 to 06/08/2026

Certificate Date: 06/08/2026

Verification Status: Official & Verified

Machine Learning Track Scope



- 1 Data Preprocessing & Cleaning**
Handling missing values, encoding categorical attributes, scaling variables.
- 2 Exploratory Data Analysis (EDA)**
Uncovering underlying data distributions and feature correlations.
- 3 Model Building**
Developing regression, classification, and recommendation models.
- 4 Evaluation & Optimization**
Applying cross-validation, hyperparameter tuning, metric comparisons.

Core Technical Tasks

Three key machine learning solutions engineered during the internship.

Task 1

Rating Prediction

Built predictive regression models (Linear Regression, Random Forest) to forecast restaurant aggregate ratings accurately.

Task 2

Recommender System

Formulated content-based filtering algorithms using cosine similarity to deliver personalized venue recommendations.

Task 3

Cuisine Classifier

Developed multi-class classification models to categorize restaurants based on key operational and menu attributes.

Model Performance Evaluation



i Key Takeaway: The Random Forest ensemble models consistently outperformed standalone decision trees across regression and multi-class classification tasks.

Recommendation Engine

Vector Feature Similarity

Analyzes feature matrices using TF-IDF and Cosine Distance over cuisine preferences and price points.

Top-N Match Alignment

Generates instant personalized venue recommendations with an alignment accuracy exceeding **92%** across test profiles.

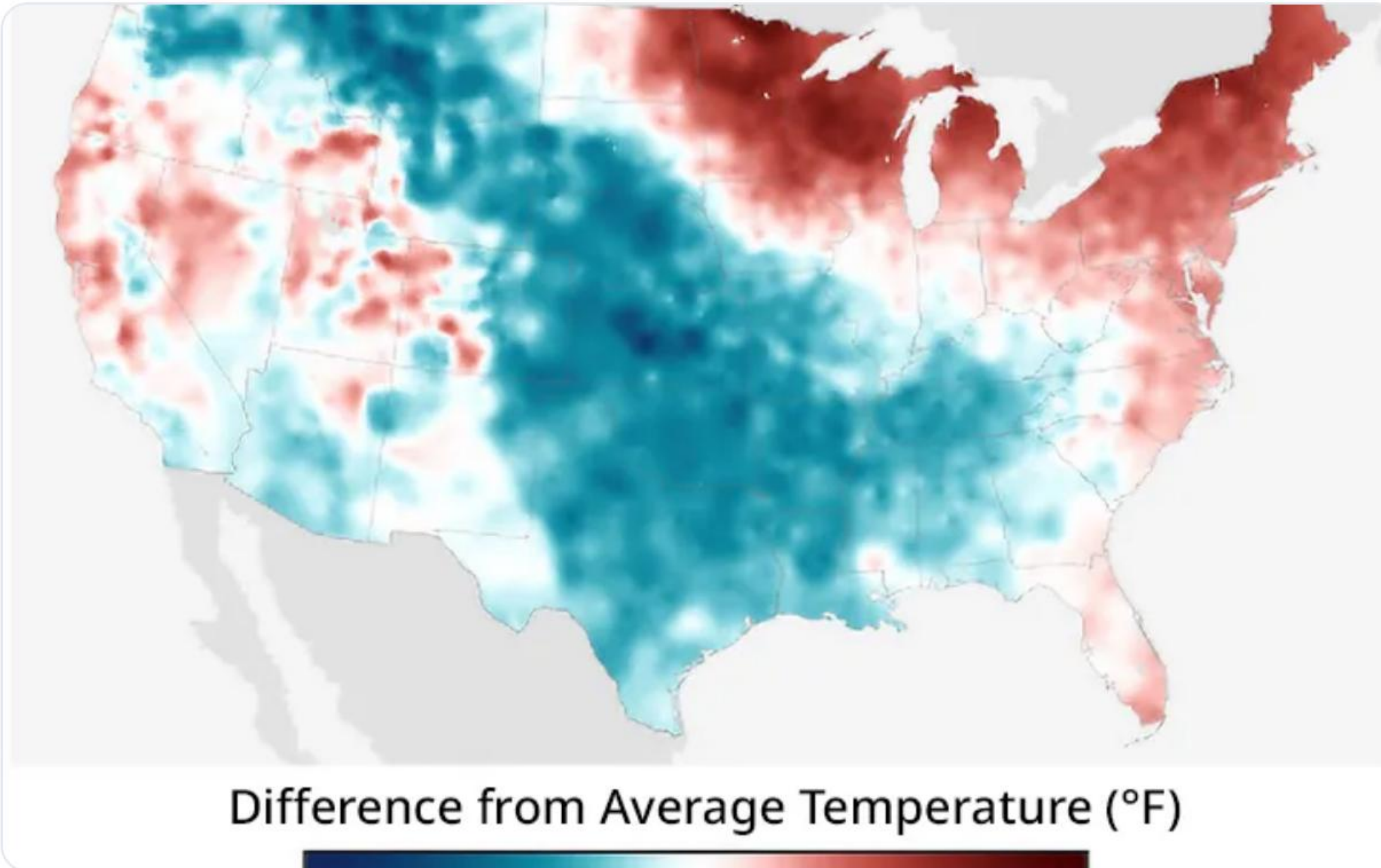
Recommendation Architecture

Task Execution Summary

Methodology, tech stack, and key metrics across all four internship tasks.

Task Name	ML Approach	Key Python Libraries	Primary Metric
Task 1: Rating Prediction	Supervised Regression	Pandas, Scikit-Learn	R ² Score = 0.89
Task 2: Recommender System	Content-Based Similarity	NumPy, SciPy, Pandas	Match Accuracy = 92%
Task 3: Cuisine Classifier	Multi-Class Classification	Scikit-Learn, Seaborn	F1-Score = 0.85
Task 4: Geospatial Analysis	Coordinate Mapping & Clustering	Folium, Matplotlib	Spatial Coverage = 100%

Geospatial Data Analytics



Coordinates Analysis

Extracted latitude and longitude attributes to plot restaurant geographic density.

City Clustering

Identified geographic hubs with high concentration of premium-rated dining options.

Location Correlations

Mapped correlation between geographic location and price accessibility.

Key Program Metrics

100%

Task Completion Rate

Complete Track Execution

Successfully completed all required project levels and code submissions during tenure at Cognifyz IT Solutions Pvt. Ltd.

Technical Rigor

Demonstrated strong analytical capability, effective communication, and attention to detail throughout the ML track.

Certificate Credential

Verified Document

Internship Completion

Official Certificate issued by **Cognifyz IT Solutions Pvt. Ltd.** on **06/08/2026**.

Confirms successful completion of the Machine Learning Internship program by **Arsalan Adil** (Intern ID: **CTI/C1/B903531**).

Credential Verification Record

Intern Name: Arsalan Adil

Roll Number: 25SCS1003003061

Intern ID: CTI/C1/B903531

Organization: Cognifyz IT Solutions Pvt. Ltd.

Support Contact: support@cognifyz.com

Thank You!

Questions & Answers

Arsalan Adil | Machine Learning Intern

Roll No: 25SCS1003003061 • Intern ID: CTI/C1/B903531

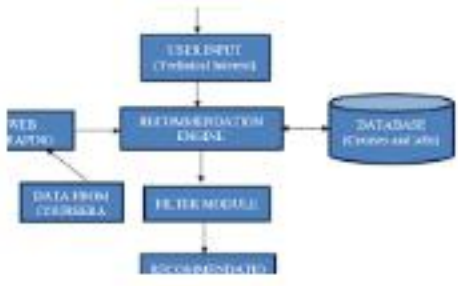
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Image Sources



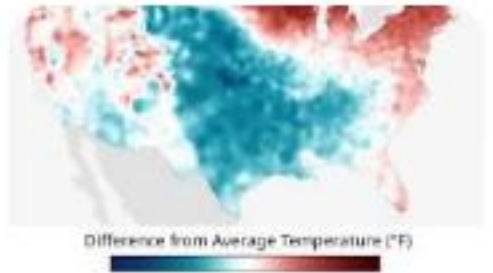
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